

8-LAYER ENTERPRISE ARCHITECTURE

LAYER 1 — DATA SOURCES (12 SYSTEMS)

Internal Guidewire Claims · Duck Creek Policy · TriZetto Medical Billing · SAP Finance · Salesforce CRM · Workday HR/Benefits · Sedgwick WC TPA
External ISO ClaimSearch · LexisNexis Risk · NICB · CMS Provider Exclusions · NOAA Weather · DMV Feeds · OFAC Sanctions · Court Records · IRS TIN

LAYER 2 — INGESTION PIPELINE (SIMULATED KAFKA)

Topics claims.fnol · claims.update · policy.change · medical.bill · repair.invoice · payment.issued · iso.claimsearch · external.enrichment
Pattern Batch (nightly 100K) + Real-time (<500ms per claim) · Same MERGE logic · Swap real Kafka in Phase 6

LAYER 3 — MULTI-MODAL AI PROCESSING

CV Pipeline Damage photos · Repair invoices · Accident scenes · Medical docs
5 CV Layers EXIF/Metadata · ELA+FFT Pixel Forensics · Duplicate Detection · Cross-Document Consistency · Temporal Consistency
OCR+LLM Entity extraction → graph loading · Anomaly flagging · Template similarity detection
Synthetic ID Discriminative detection first → generative detection capability designed in

LAYER 4 — ENTITY RESOLUTION

Cross-system Master person_id across Guidewire + Salesforce + Workday + Sedgwick
Fuzzy match Names · Phones · Addresses · DOB · SSN-last4 · Device fingerprint
ISO match Cross-carrier identity resolution via ISO ClaimSearch xref

LAYER 5 — GRAPH STORE (NEO4J AURADB)

24 node labels Claim · ClaimVersion · CoverageLine · ClaimPayment · Person · Employee · Dependent · Policy · Vehicle · Fleet · MedicalProvider · MedicalBill · RepairShop · Attorney · TowCompany · Employer · Address · Phone · CrossSystemID · Adjuster · BankAccount · LitigationEvent · InjuryRecord · FraudRing
Temporal Properties + timestamps on all nodes · GDS temporal algorithms · Time-series graph for complex patterns (Phase 2)

LAYER 6 — INTELLIGENCE ENGINE

GDS Connected Components · Louvain Community · PageRank · Betweenness Centrality · Temporal WCC
GNN PyTorch Geometric designed in · Optional activation · Heterogeneous graph neural network for collusion
Scoring 21 fraud patterns · LOB-specific weights · Payment hold logic · Cross-carrier signals from ISO

LAYER 7 — AGENTIC AI (HUMAN-IN-THE-LOOP)

Co-pilot AI assembles evidence package · Suggests investigation steps · Drafts SIU referral narrative

Human Analyst reviews · Approves/rejects referral · Adds external evidence · Closes case

Autonomous Low-risk only: duplicate detection · OFAC screening · Payment hold triggers

LAYER 8 — INVESTIGATOR DASHBOARD

Views Graph ring visualization · Temporal timeline · Geospatial map · Adjuster workqueue · Cross-carrier hits

Actions SIU referral · Payment hold · Watchlist flag · Case notes · Evidence upload

21 FRAUD PATTERNS ACROSS 6 CATEGORIES**Cat 1 — Healthcare/Medical**

HIGH Medical mill (1 doctor → 50+ claims)

HIGH CPT upcoding (billed vs allowed ratio)

HIGH Phantom treatment (no facility record)

MED Duplicate billing across carriers

MED Unbundling CPT codes

LOW Exaggerated soft tissue injury

Cat 3 — Temporal/Sequence

HIGH Claim velocity (3+ claims in 90 days)

HIGH Impossible timeline (treatment before incident)

MED Reserve pattern anomaly (rapid escalation)

MED Policy lapse/reinstate cycle before claim

LOW Seasonal fraud clustering

Cat 5 — Insider/Adjuster

HIGH Adjuster + shop collusion (approval rate anomaly)

HIGH Adjuster + attorney pipeline (settlement inflation)

MED Phantom payment (payee = adjuster network)

MED Reopened closed claims pattern

Cat 2 — Property-Specific

HIGH Arson ring (multi-property, same owner/agent)

HIGH Flood fraud (pre-existing damage claimed)

MED Contractor collusion (inflated estimates)

MED Contents inflation (no purchase records)

LOW Post-loss policy inception

Cat 4 — Identity & Synthetic

HIGH Synthetic identity (thin credit file + new SSN)

HIGH Identity takeover (address/phone sudden change)

MED Phantom passenger recycling

LOW Address fraud (rate territory mismatch)

Cat 6 — Cross-Carrier (ISO)

HIGH Same incident filed at 2+ carriers

HIGH Serial claimant across carriers (ISO hit)

MED Provider excluded at one carrier still billing another

MED Cross-carrier medical mill network

BUILD PHASES — PRODUCTION ROADMAP

PHASE	DELIVERABLE	KEY COMPONENTS	AURADB NODES
A — Schema	Production Neo4j schema	24 node labels · 32 relationships · all constraints · temporal indexes	0 (schema only)
B — Data Generator	50K node production dataset	5 source systems · 21 fraud patterns · 10 years · Hartford geography · ISO simulation	~50K nodes
C — Entity Resolution	Cross-system ID pipeline	Fuzzy match · master ID · CrossSystemID nodes · ISO xref	+5K xref nodes
D — Intelligence Engine	GDS + scoring all 21 patterns	Temporal GDS · LOB weights · payment hold · ISO signal · GNN design-in	+score properties
E — Multi-Modal AI	CV + OCR + LLM pipeline	5-layer CV · entity extraction · anomaly detection · synthetic ID discriminator	+image metadata nodes
F — Agentic Co-pilot	Human-in-loop investigator	Evidence assembly · SIU narrative · analyst approval workflow	+case nodes
G — Dashboard v2	Production investigator UI	Timeline · geo map · cross-carrier view · adjuster queue · evidence panel	read only
H — GNN (Optional)	PyTorch Geometric collusion	Heterogeneous GNN · trained on labeled rings · replaces/augments GDS Louvain	read only